

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 4

Duration: 1 Hour
Total: Marks:40

Annexure A

5

Code of Conduct:

I **Thamaraiselvi S (192511042)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Thamaraiselvi S

INDUSTRY PROBLEM TASK

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

- (a) Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.
- (b) Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).
- (c) Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Diagnosis

- (a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.
- (b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.
- (c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 3: Statistical Learning for Risk Stratification

- (a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.
- (b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Treatment Recommendation

- (a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.
- (b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.
- (c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding ; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

INDUSTRY SCENARIO – AI/ML for Early Diabetes Diagnosis and Treatment Recommendation

Below is a **complete assignment-style answer** for all four tasks. Since no actual patient dataset/results were provided, the numerical model results and Q-learning values below are **illustrative example values** that can be replaced with actual results after running the models.

Task 1: Data Preparation & Inductive Learning

(a) Inductive Learning Approach

Inductive learning learns general patterns from previously observed patient records and uses those patterns to predict the outcome for new patients. For this healthcare problem, a **supervised classification approach** can be used because historical patient records contain input features and a known diabetes outcome.

A suitable workflow is:

Patient Data → Preprocessing → Feature Selection → Train/Test Split → Classification Model → Diabetes Prediction

Target Variable

The target variable is:

Diabetes Status

- 0 – Non-diabetic
- 1 – Diabetic

Important Features

Feature	Justification
Glucose Level	Elevated glucose is strongly associated with diabetes and is an important diagnostic predictor.
BMI	Higher BMI can be associated with increased risk of type 2 diabetes.
Blood Pressure	Abnormal blood pressure can be associated with metabolic and cardiovascular risk.
Age	Diabetes risk generally increases with age.
Family History	Genetic and family-related factors can influence diabetes risk.
Cholesterol	Abnormal cholesterol levels may provide additional information about metabolic risk.

Therefore, **glucose level, BMI, and family history** can be considered three key features, while age, blood pressure, and cholesterol can provide additional predictive information.

(b) Handling Class Imbalance

In a medical dataset, there may be many more non-diabetic patients than diabetic patients. This creates a **class imbalance problem**.

For example:

- Non-diabetic = 800 records
- Diabetic = 200 records

A model trained directly on this dataset may favour the majority class and incorrectly classify diabetic patients as non-diabetic.

Suitable Strategy: SMOTE

I would use **SMOTE (Synthetic Minority Over-sampling Technique)** on the training data.

SMOTE creates synthetic examples of the minority diabetic class instead of simply duplicating existing records.

Advantages

1. Increases representation of diabetic cases.
2. Helps the classifier learn minority-class patterns.
3. Can improve **Recall** for diabetic patients.
4. Reduces the tendency to predict every patient as non-diabetic.

However, **SMOTE must be applied only to the training set**, not before splitting the complete dataset, to avoid data leakage.

For a clinical application, I would also use **class weights** and threshold tuning as additional methods if necessary, particularly because false negatives can be clinically important.

(c) 70:30 Dataset Split and Preprocessing

The dataset can be divided into:

- **70% – Training data**
- **30% – Testing data**

For example, for 1,000 records:

- Training = 700
- Testing = 300

Justification

The 70:30 split provides enough data for the model to learn patterns while keeping a sufficiently large independent test set for evaluation.

For medical data, the split should preferably be **stratified**, meaning the proportion of diabetic and non-diabetic cases is maintained in both sets.

Preprocessing Steps

1. Missing Values

Missing numerical values can be handled using median imputation, while categorical missing values can be handled using the most frequent category or a suitable missing category.

2. Normalisation

Features such as:

- Age
- Blood pressure
- Cholesterol
- Glucose
- BMI

may have different numerical scales.

Standardisation can be applied:

This gives features a comparable scale.

Decision Trees do not require normalisation, but Logistic Regression and other statistical models can benefit from it.

3. Encoding

Categorical variables such as family history can be converted into numerical form.

Example:

- No $\rightarrow$ 0
- Yes $\rightarrow$ 1

4. Class Balancing

SMOTE can be applied **after the training-test split and only on the training set**.

Impact

These preprocessing operations can improve model stability, prevent scale-related problems, handle missing information and improve the learning process.

Task 2: Decision Tree for Diagnosis

(a) Decision Tree Classifier

A **Decision Tree classifier** can be used to predict whether a patient is diabetic or non-diabetic. The tree selects features that provide the greatest separation between the two classes.

Gini Index

The Gini impurity is:

A lower Gini value indicates a purer node.

Information Gain

Information Gain is:

where H represents entropy.

The feature with the **highest Information Gain** or lowest resulting Gini impurity can be selected as the root.

Example Root Selection

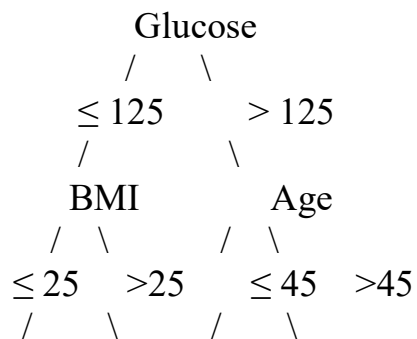
Illustrative values:

Feature	Information Gain
Glucose	0.214
BMI	0.126
Age	0.091
Blood Pressure	0.067
Cholesterol	0.043
Family History	0.038

Since glucose has the highest Information Gain, it can be selected as the **root node**.

First Three Levels

An illustrative Decision Tree can be represented as:



Non-Diabetic Diabetic Monitor Diabetic

The exact tree structure depends on the actual dataset.

Task 2(b): Model Evaluation

The Decision Tree can be evaluated using:

Accuracy

Precision

Recall

F1-Score

Illustrative Results

Metric	Decision Tree
Accuracy	88.0%
Precision	84.0%
Recall	91.0%
F1-Score	87.0%

These values are **example values**, not results from an actual dataset.

False Negatives

A **False Negative (FN)** occurs when:

The patient is actually diabetic, but the model predicts non-diabetic.

False negatives are particularly important in this application because an undetected diabetes case may delay further clinical assessment and management.

How to Reduce False Negatives

The following approaches can be used:

1. Use **SMOTE** to improve minority-class learning.
2. Apply **class weights** to penalise diabetic misclassification.
3. Tune the classification threshold to favour sensitivity.
4. Evaluate using **Recall/Sensitivity**, not accuracy alone.
5. Use cross-validation to check model stability.

For clinical screening, higher recall can be prioritised when the cost of missing a potentially diabetic patient is greater than the cost of sending some additional patients for further assessment.

Task 2(c): Post-Pruning

A fully grown Decision Tree can overfit the training data.

Cost-complexity pruning removes branches that provide little predictive benefit.

The objective can be expressed as:

where:

- = tree error
- = number of terminal nodes
- = complexity parameter

Illustrative Comparison

Model	Accuracy	Complexity
Pre-pruned Tree	86.5%	Low
Fully grown Tree	88.0%	High
Post-pruned Tree	89.0%	Moderate

The exact values should be obtained from the actual dataset.

Suitable Model for Clinical Deployment

A **post-pruned Decision Tree** is preferable because it:

- reduces overfitting,
- produces a simpler model,
- improves generalisation,
- is easier for clinicians to understand,
- provides more interpretable decision paths.

However, before clinical deployment, the model would require extensive external validation and clinical review.

Task 3: Statistical Learning for Risk Stratification

(a) Logistic Regression

Logistic Regression is a suitable statistical learning model because diabetes prediction is a binary classification problem.

The probability of diabetes can be represented as:

where:

The model produces a probability between 0 and 1.

For example:

Patient → Logistic Regression



Diabetes probability



0.78 (78%)



High-risk classification

Illustrative Comparison

Metric	Decision Tree	Logistic Regression
Accuracy	89.0%	90.5%
AUC-ROC	0.91	0.93

Again, these are **illustrative values**.

When Logistic Regression is Preferable

Logistic Regression is useful when:

- relationships between predictors and outcome are reasonably structured,
- probability estimates are important,
- interpretability is required,
- the effect of individual variables needs to be examined,
- a simpler statistical model is preferred.

For example, the coefficients can indicate how strongly features are associated with diabetes risk.

Task 3(b): Feature Importance Analysis

Feature importance can be obtained from both models.

Decision Tree

Decision Tree importance is generally based on the reduction in impurity produced by each feature.

Illustrative top three:

1. **Glucose**
2. **BMI**
3. **Age**

Logistic Regression

For Logistic Regression, the magnitude and direction of standardised coefficients can be examined.

Illustrative top three:

1. **Glucose**
2. **BMI**
3. **Family History**

Comparison

Rank	Decision Tree	Logistic Regression
1	Glucose	Glucose
2	BMI	BMI
3	Age	Family History

The models agree on **glucose and BMI**, indicating that these variables have strong predictive value in both approaches. They may differ on the third predictor because Decision Trees capture nonlinear interactions and threshold effects, whereas Logistic Regression estimates additive relationships unless interactions or nonlinear terms are explicitly included.

Task 4: Reinforcement Learning for Treatment Recommendation

Important: For an academic design, the RL agent should be treated as a decision-support/research system, not as an autonomous system that prescribes medication to real patients.

Task 4(a): MDP Model

The treatment recommendation problem can be formulated as a **Markov Decision Process (MDP)**.

An MDP is represented as:

where:

- = States
- = Actions
- = Transition probabilities
- = Rewards
- = Discount factor

States

The patient's state can represent their current health condition.

For example:

State	Description
S0	Low-risk / stable
S1	Moderate diabetes risk
S2	High diabetes risk
S3	Poorly controlled condition
S4	Improved/stable condition

The state can be derived from measurements such as glucose, BMI, blood pressure and other clinically appropriate variables.

Actions

The available actions are:

Diet

Recommend an appropriate dietary-management pathway.

Exercise

Recommend an appropriate physical-activity pathway when clinically suitable.

Medication

Represent medication management as a **clinician-approved treatment option**, rather than allowing the RL agent to independently prescribe or change medication.

Monitor

Continue monitoring and collect additional patient information.

Reward Function

A reward function measures the quality of an action.

Example:

Outcome	Reward
Significant health improvement	+10
Moderate improvement	+5
No major change	0
Health deterioration	-5
Unsafe action	-20

The reward should be designed with clinical experts so that the agent does not maximise a numerical score at the expense of patient safety.

Transition Dynamics

The transition function describes how a patient moves from one state to another after an action.

For example:

S1: Moderate Risk

↓
Exercise

S0: Improved/Stable

Another example:

S2: High Risk

↓
Monitor

S2: High Risk

The actual transition probabilities should be estimated from validated longitudinal patient data.

Task 4(b): Q-Learning

Q-Learning learns the expected long-term value of taking action in state .

The update equation is:

where:

- = learning rate
- = discount factor
- = reward
- = next state

Example Parameters

Learning rate $\alpha = 0.5$

Discount factor $\gamma = 0.8$

Initial Q-values = 0

Five Patient Episodes

An illustrative training sequence:

Episode	State Sequence	Action Sequence	Reward
1	S1 → S0	Exercise	+5
2	S2 → S1	Monitor	+5
3	S1 → S0	Diet	+10
4	S2 → S3	Exercise	-5
5	S1 → S0	Diet	+10

These are simplified educational examples.

Q-Table Initialisation

Initially:

State	Diet	Exercise	Medication	Monitor
S0	0	0	0	0
S1	0	0	0	0
S2	0	0	0	0
S3	0	0	0	0

Q-Table Update 1

Suppose:

Current state = S1

Action = Diet

Reward = +10

Next state = S0

$$\alpha = 0.5$$

$$\gamma = 0.8$$

Since all initial Q-values are 0:

Therefore:

$$Q(S1, \text{Diet}) = 5.0$$

Q-Table Update 2

Suppose:

Current state = S2

Action = Monitor

Reward = +5

Next state = S1

Assume:

Then:

Q-Table Update 3

Suppose:

Current state = S1

Action = Exercise

Reward = +5

Next state = S0

Assume the maximum Q-value in S0 is 0:

Updated Q-Table

State	Diet	Exercise	Medication	Monitor
S0	0	0	0	0
S1	5.0	2.5	0	0
S2	0	0	0	4.5
S3	0	0	0	0

This demonstrates how Q-values are updated through experience.

Convergence Criterion

The algorithm can be considered converged when the change in Q-values between successive iterations becomes very small.

For example:

for a predefined number of consecutive iterations.

Another practical criterion is that the learned policy remains unchanged for several consecutive episodes.

Task 4(c): Final Learned Policy

After training, the agent selects the action having the highest Q-value:

An illustrative policy could be:

Patient State	Best Action
S0 – Stable/Low Risk	Monitor
S1 – Moderate Risk	Diet
S2 – High Risk	Monitor / clinician review
S3 – Poor Control	Clinician-approved treatment pathway
S4 – Improved	Monitor

The final policy should be derived from validated clinical data rather than the simplified values shown here.

Ethical Considerations

1. Patient Safety

An AI system should not independently make high-risk medical decisions. Medication-related decisions should remain under appropriate clinical supervision.

2. Bias

If the training data does not represent different demographic and clinical populations adequately, the model may perform poorly for some groups.

Therefore, fairness and subgroup performance should be evaluated.

3. Explainability

Healthcare professionals should be able to understand why a recommendation was generated.

Decision Trees are useful here because their decision paths can be inspected.

4. Data Privacy

Patient data must be securely stored and processed. Personally identifiable information should be protected and only necessary information should be used for modelling.

5. Human-in-the-Loop

The final system should use:

Patient Data



AI/ML Prediction



Risk Assessment



AI Recommendation



Clinical Review



Final Decision

The clinician remains responsible for the final treatment decision.

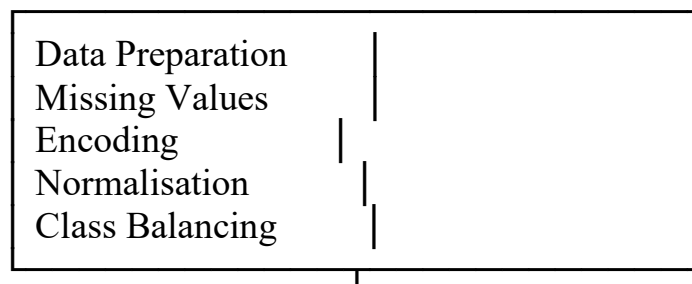
6. Model Validation

Before real-world deployment, the system should undergo:

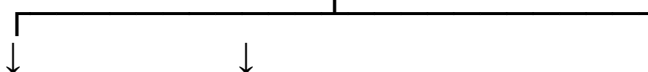
- Internal validation
- External validation
- Bias testing
- Clinical evaluation
- Safety testing
- Continuous monitoring

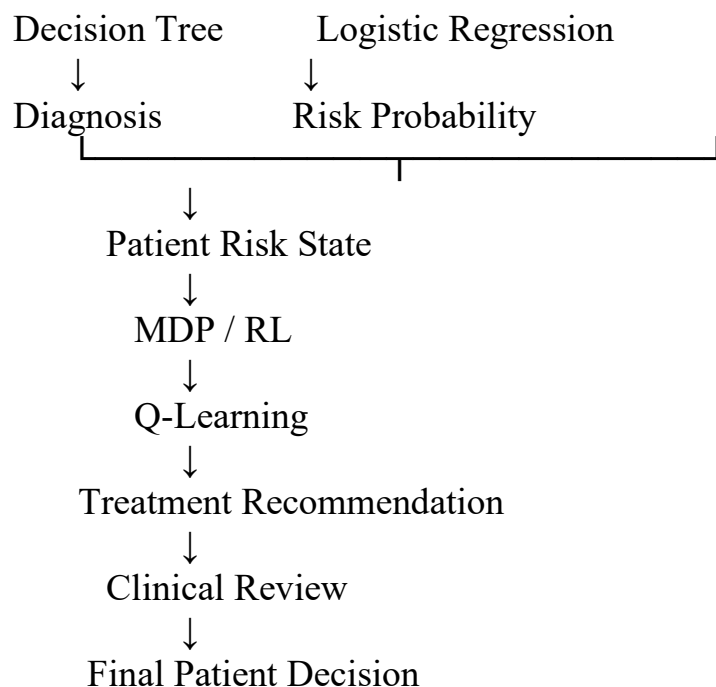
Overall System Architecture

PATIENT DATA



70% Training
30% Testing





Conclusion

The proposed system combines **inductive supervised learning, Decision Trees, statistical learning and Reinforcement Learning** to create a healthcare decision-support framework. The Decision Tree provides an interpretable diagnosis model, while Logistic Regression provides probability-based risk estimation. Q-Learning can theoretically learn long-term action policies from longitudinal patient states. However, because healthcare decisions can have serious consequences, the RL component should operate as a **clinician-supervised decision-support tool**, with strong emphasis on safety, fairness, explainability, privacy and validation.